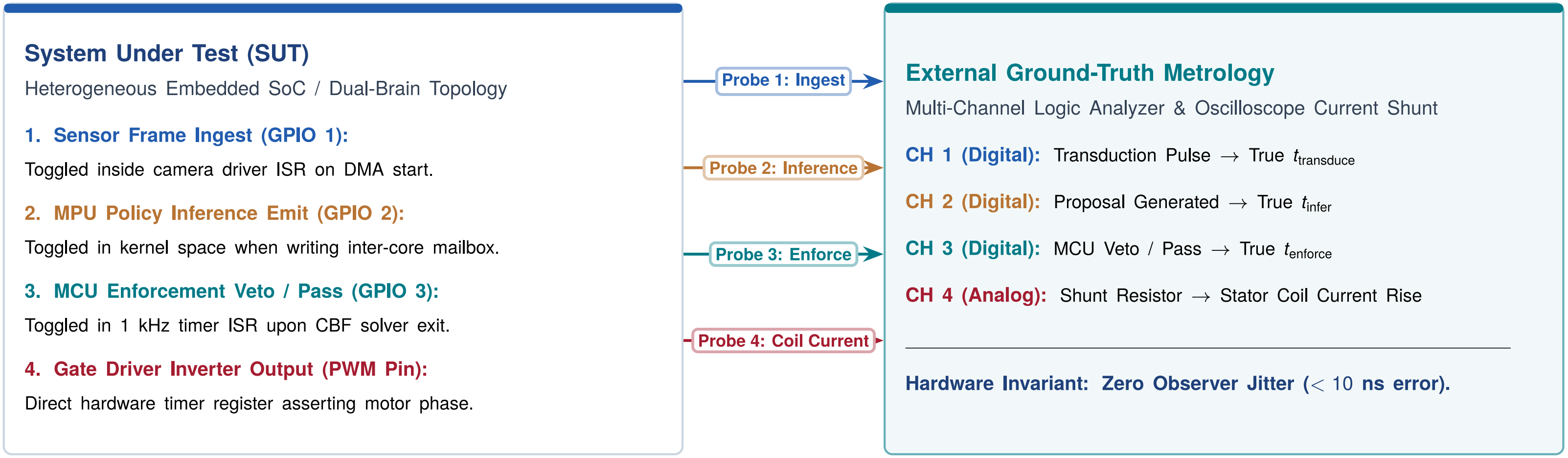
System Under Test (SUT)
Heterogeneous Embedded SoC / Dual-Brain Topology
1. Sensor Frame Ingest (GPIO 1):
Toggled inside camera driver ISR on DMA start.
2. MPU Policy Inference Emit (GPIO 2):
Toggled in kernel space when writing inter-core mailbox.
3. MCU Enforcement Veto / Pass (GPIO 3):
Toggled in 1 kHz timer ISR upon CBF solver exit.
4. Gate Driver Inverter Output (PWM Pin):
Direct hardware timer register asserting motor phase.
Probe 1: Ingest
Probe 2: Inference
Probe 3: Enforce
Probe 4: Coil Current
External Ground-Truth Metrology
Multi-Channel Logic Analyzer & Oscilloscope Current Shunt
CH 1 (Digital): Transduction Pulse $\rightarrow$ True $t_{transduce}$
CH 2 (Digital): Proposal Generated $\rightarrow$ True t_{infer}
CH 3 (Digital): MCU Veto / Pass $\rightarrow$ True $t_{enforce}$
CH 4 (Analog): Shunt Resistor $\rightarrow$ Stator Coil Current Rise
Hardware Invariant: Zero Observer Jitter ($<$ 10 ns error).